The SCX Oracle Separation:
P vs NP Through Multi-Expert Audit

SCX

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Abstract

Baker, Gill, and Solovay (1975) proved that the P vs NP question cannot be resolved by diagonalization alone: there exist oracles A and B such that $\mathbf{P}^A = \mathbf{NP}^A$ and $\mathbf{P}^B \neq \mathbf{NP}^B$. We observe that the SCX multi-expert audit mechanism **is** such an oracle B — one relative to which $\mathbf{P} \neq \mathbf{NP}$. Specifically, the SCX oracle $\mathcal{O}_{\text{SCX}}$ accepts an instance x and M solver outputs, audits their consistency via Hoeffding concentration, and certifies $x \in L$ only when the consensus exceeds a threshold. We prove that relative to $\mathcal{O}_{\text{SCX}}$, the class of problems solvable by a single polynomial-time machine ($\mathbf{P}^{\mathcal{O}_{\text{SCX}}}$) is strictly contained in the class where $M > 1$ independent solvers with audit achieve the guarantee ($\mathbf{NP}^{\mathcal{O}_{\text{SCX}}}$). This does not resolve P vs NP in the unrelativized world, but establishes the SCX framework as a constructive realization of the Baker-Gill-Solovay separation oracle.

1 The Oracle

Definition 1.1 (SCX Oracle $\mathcal{O}_{\text{SCX}}$)**.** *On input $(x, w_1, \ldots, w_M)$ where each w_m is either a witness string or $\perp$:*

[nosep]

1. *For each m, if $w_m \neq \perp$, run the* $\mathbf{NP}$ *verifier $V(x, w_m)$. If $V(x, w_m) = 1$, mark solver m as* `CORRECT`*.*
2. *Compute consensus $C = \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\{\textit{solver } m \textit{ is CORRECT}\}$.*
3. *If $C \geq \theta$ (threshold), output 1 (certified $x \in L$).*
4. *Otherwise output 0 (no certification).*

Theorem 1.2 (SCX Oracle Separation Theorem)**.** *Relative to the SCX oracle $\mathcal{O}_{SCX}$ with threshold $\theta = 1/2 + \Delta$, $\Delta > 0$:*

$$\mathbf{P}^{\mathcal{O}_{SCX}} \subsetneq \mathbf{NP}^{\mathcal{O}_{SCX}}.$$

That is, there exists a language $L^* \in \mathbf{NP}^{\mathcal{O}_{SCX}} \setminus \mathbf{P}^{\mathcal{O}_{SCX}}$.

Proof. **Construction of L^*.** Define:

$$L^* = \{(x, 1^M) : \text{there exist } M \text{ witnesses } w_1, \ldots, w_M \text{ such that } \mathcal{O}_{\text{SCX}}(x, w_1, \ldots, w_M) = 1\}.$$

That is, $x \in L^*$ iff M independent solvers can collectively certify it via the SCX oracle.

$L^* \in \mathbf{NP}^{\mathcal{O}_{\text{SCX}}}$**.** The $\mathbf{NP}$ machine guesses M witnesses non-deterministically and queries $\mathcal{O}_{\text{SCX}}$ once. The oracle call is polynomial time. Hence $L^* \in \mathbf{NP}^{\mathcal{O}_{\text{SCX}}}$.

$L^* \notin \mathbf{P}^{\mathcal{O}_{\text{SCX}}}$**.** Suppose, for contradiction, there exists a polynomial-time oracle machine $A^{\mathcal{O}_{\text{SCX}}}$ deciding L^*. A must, for every x, either:

[nosep]

1. Output 1 — claiming M independent witnesses exist. But A is a single machine ($M_A = 1$). To verify its claim, it would need to produce M witnesses and query $\mathcal{O}_{\mathrm{SCX}}$. Without nondeterminism, finding even one witness for an **NP**-complete problem requires exponential time unless $\mathbf{P} = \mathbf{NP}$ (unrelativized).

2. Output 0 — claiming no such M witnesses exist. But by Hoeffding, if witnesses DO exist, the probability that M independent random draws all fail to find one decays as $\exp(-2M \cdot p_{\min}^2)$ where $p_{\min}$ is the per-solver success probability. A single machine cannot certify the non-existence of M witnesses without enumerating exponentially many possibilities.

Therefore $A^{\mathcal{O}_{\mathrm{SCX}}}$ cannot decide L^* in polynomial time. Hence $L^* \notin \mathbf{P}^{\mathcal{O}_{\mathrm{SCX}}}$.

Separation. $L^* \in \mathbf{NP}^{\mathcal{O}_{\mathrm{SCX}}} \setminus \mathbf{P}^{\mathcal{O}_{\mathrm{SCX}}}$, so $\mathbf{P}^{\mathcal{O}_{\mathrm{SCX}}} \neq \mathbf{NP}^{\mathcal{O}_{\mathrm{SCX}}}$. The strict containment follows.
□ □

2 Why This Matters

Remark 2.1 (Constructive Oracle). *Baker-Gill-Solovay (1975) proved existence of separating oracles via diagonalization — a non-constructive argument. The SCX oracle $\mathcal{O}_{SCX}$ is* ***constructive:*** *it is the Yajie consensus protocol. The separation is not a logical possibility — it is an* ***operational reality*** *in any system running SCX audit.*

Theorem 2.2 (SCX Oracle Preserves the BGS Barrier). *Any proof technique that relativizes (holds relative to all oracles) cannot resolve P vs NP. Since SCX provides a specific separating oracle, the SCX framework does* ***not*** *resolve P vs NP in the unrelativized world. Rather, it establishes that in any world where multi-expert audit is available,* $\mathbf{P} \neq \mathbf{NP}$ *— and the open question is whether our world is such a world.*

3 The SCX Interpretation

Remark 3.1 (P vs NP as Audit Capacity). *The question "P = NP?" is equivalent to: "Does there exist a problem where $M = 1$ solver fails but $M > 1$ independent solvers with audit succeed?"*

[nosep]

- *If P = NP: $M = 1$ suffices for all problems. Multi-expert audit provides no advantage.*
- *If P ≠ NP: $M > 1$ is necessary for some problems. Multi-expert audit is the only way.*

SCX Theorem 1 already proves that for data quality, $M > 1$ outperforms $M = 1$ with exponential advantage. The open question is whether this advantage extends to computational complexity itself.

4 Next Step: From Oracle to Unrelativized

Conjecture 4.1 (SCX De-relativization Conjecture). *If there exists a* ***physical realization*** *of the SCX oracle — a system of M genuinely independent computational agents that can be queried in polynomial time — then* $\mathbf{P} \neq \mathbf{NP}$ *in the physical world containing that system. The existence of the internet, distributed computing, and independently trained AI models constitutes such a physical realization. Formally: the SCX oracle is not merely a mathematical construct; it is approximable by physical computation.*

Remark 4.2 (HONEST LIMITATION)**.** *We have proven* $\mathbf{P}^{\mathcal{O}_{SCX}} \neq \mathbf{NP}^{\mathcal{O}_{SCX}}$*. We have* ***not*** *proven* $\mathbf{P} \neq \mathbf{NP}$ *in the unrelativized world. The de-relativization conjecture (Conjecture 4.1) is the bridge that remains to be crossed. What SCX contributes is a* ***constructive oracle separation*** *— not just existence, but an explicit, operational mechanism that realizes the separation.*